Assignment 10: Multiple Choice Questions on Ensemble Learning

1. What is the main advantage of using an ensemble of predictors?
A) It reduces the training time significantly
B) It improves the predictive performance compared to individual models
C) It eliminates the need for data preprocessing
D) It simplifies the model interpretation
Answer: B) It improves the predictive performance compared to individual models
Explanation: Ensemble methods combine multiple weak learners to create a strong learner, often achieving better accuracy than the best individual model.

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2. In ensemble learning, what is the function of the aggregation step?
A) Selecting the best performing model from the ensemble
B) Combining the predictions of multiple models to make a final decision
C) Training multiple models in parallel
D) Reducing the number of models in the ensemble
Answer: B) Combining the predictions of multiple models to make a final decision
Explanation: Aggregation is crucial in ensemble learning as it combines outputs from different models using methods like voting (for classification) or averaging (for regression).

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3. Which of the following is true about a Hard Voting Classifier?
A) It selects the prediction with the highest confidence score
B) It assigns different weights to different classifiers
C) It predicts the class that receives the majority vote from all classifiers
D) It always performs better than a single classifier
Answer: C) It predicts the class that receives the majority vote from all classifiers
Explanation: A hard voting classifier aggregates votes from multiple models and selects the class with the most votes, without considering confidence scores.

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4. What differentiates Soft Voting from Hard Voting?
A) Soft voting uses probability estimates, while hard voting counts votes
B) Hard voting requires probability estimation, while soft voting does not
C) Hard voting gives different weights to classifiers, while soft voting does not
D) Soft voting works only with regression models
Answer: A) Soft voting uses probability estimates, while hard voting counts votes
Explanation: Soft voting averages the predicted probabilities and selects the class with the highest probability, making it often more accurate than hard voting.

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5. What is the key difference between Bagging and Pasting?

A) Bagging trains each predictor on different training algorithms, while pasting uses the same algorithm
B) Bagging samples with replacement, while pasting samples without replacement
C) Pasting is more computationally expensive than bagging
D) Bagging uses more predictors than pasting
Answer: B) Bagging samples with replacement, while pasting samples without replacement
Explanation: In bagging, training data subsets are selected with replacement, meaning some samples may be repeated, whereas pasting does not allow repetition.

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6. Why does Bagging reduce variance?
A) It increases model complexity
B) It averages multiple predictions, reducing overfitting
C) It focuses on difficult-to-predict samples
D) It uses weak learners sequentially
Answer: B) It averages multiple predictions, reducing overfitting
Explanation: Bagging combines multiple predictors trained on different subsets, leading to lower variance while maintaining similar bias.

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7. What is the purpose of Out-of-Bag (OOB) evaluation?
A) To validate models without a separate validation set
B) To reduce computational costs during training
C) To ensure all instances are used in every training subset
D) To select the best performing model in the ensemble
Answer: A) To validate models without a separate validation set
Explanation: OOB evaluation uses samples that were not selected during bagging to assess model performance, eliminating the need for a separate validation set.

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8. How does Random Forest introduce randomness compared to traditional Decision Trees?
A) It uses a different feature set for each tree when splitting nodes
B) It uses gradient boosting instead of bagging
C) It combines weak learners in a sequential manner
D) It selects the best feature for every split in all trees
Answer: A) It uses a different feature set for each tree when splitting nodes
Explanation: Random Forest improves diversity by choosing a random subset of features at each split, making it less prone to overfitting.

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9. What is a key difference between Extra-Trees and Random Forests?
A) Extra-Trees use a fixed number of trees, while Random Forests do not
B) Extra-Trees select random thresholds for splits, while Random Forests find the best thresholds
C) Random Forests require boosting, while Extra-Trees do not
D) Extra-Trees always use more training data than Random Forests
Answer: B) Extra-Trees select random thresholds for splits, while Random Forests find the best thresholds
Explanation: Extra-Trees increase randomness by selecting thresholds randomly, which speeds up training and reduces variance.

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10. What is the main disadvantage of Boosting methods like AdaBoost?
A) They cannot be used with Decision Trees
B) They are not suitable for high-dimensional datasets
C) They require sequential training, making parallelization difficult
D) They do not work well with weak learners
Answer: C) They require sequential training, making parallelization difficult
Explanation: Boosting builds models sequentially, where each model corrects errors from the previous one, making it hard to parallelize.

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11. How does AdaBoost focus on difficult training samples?
A) It removes easy samples from training
B) It assigns higher weights to misclassified samples
C) It uses different base models for different samples
D) It reduces variance by averaging predictions
Answer: B) It assigns higher weights to misclassified samples
Explanation: AdaBoost increases the weight of misclassified samples in subsequent iterations, making the next model focus more on difficult cases.

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12. In Gradient Boosting, how does each new predictor improve upon the previous one?
A) By adjusting feature selection in each iteration
B) By learning from residual errors of the previous predictor
C) By training on new subsets of data in each iteration
D) By using only the most confident predictions
Answer: B) By learning from residual errors of the previous predictor
Explanation: Gradient Boosting trains each new model on the errors made by the previous ones, gradually reducing overall prediction error.

1. Suppose you have a dataset with 10,000 samples. If you train a Random Forest with bootstrapping, approximately how many unique samples will be included in each bootstrap sample?
   a) 10,000
   b) 7,000
   c) 6,300
   d) 5,000
   Answer: c) 6,300 (Since each sample has a $1-1/e \approx 63\%$ probability of being included in the bootstrap sample).
   Explanation: When training a Random Forest with bootstrapping, each bootstrap sample will typically include around 63% of the unique samples from the original dataset.

2. How does Out-of-Bag (OOB) error estimation work in Bagging?
   a) It computes error using a separate validation set
   b) It tests each tree on the bootstrap samples it was trained on
   c) It evaluates each tree using data that was not included in its training set
   d) It uses cross-validation for performance measurement
   Answer: c) It evaluates each tree using data that was not included in its training set
   Explanation: In bagging, Out-of-Bag (OOB) error estimation leverages the "out-of-bag" samples (those not used in a particular tree's training) to estimate the model's generalization error without needing a separate validation set.

3. A Random Forest regression model has an R-squared (R^2) value of 0.85. What percentage of variance in the data is not explained by the model?
   a) 15%
   b) 50%
   c) 85%
   d) 100%
   Answer: a) 15% (Since unexplained variance = $1-R^2=1-0.85=0.15$).
   Explanation: R^2 represents the proportion of variance in the dependent variable explained by the model. The total variance is considered to be 100%. The unexplained variance will be ($1- R^2$).

4. How does Boosting improve model performance?
   a) By training multiple weak learners independently
   b) By sequentially adjusting weights of misclassified samples
   c) By randomly selecting features to reduce overfitting
   d) By reducing variance through averaging predictions
   Answer: b) By sequentially adjusting weights of misclassified samples
   Explanation: Boosting improves model performance by sequentially combining multiple "weak learners" (models that perform only slightly better than random guessing) into a strong learner, focusing on correcting errors made by previous models, and ultimately leading to more accurate and robust predictions.

5. Which of the following is not related to boosting?
   a) Non uniform distribution

b) Re-weighting
c) Re-sampling
d) Sequential style
Answer: c) Re-sampling
Explanation: Re-sampling is done with the bagging technique. Boosting uses a non-uniform distribution, during the training the distribution will be modified and difficult samples will have higher probability. And it follows a sequential style to generate complementary base-learners by re-weighting the learner.

6. A deep learning model has 10,000,000 parameters, and each parameter update takes 0.0001 seconds. If you train for 50 epochs on 1,000,000 samples, how long will training take?
a) 500 seconds
b) 5,000 seconds
c) 50,000 seconds
d) 500,000 seconds
Answer: c) 50,000 seconds
Explanation: training time for a machine learning model (given with parameters, updating time, epochs).
Then the total time training will take = Number of parameters * Updating time * Number of epochs.
Total time = $10{,}000{,}000 \times 0.0001 \times 50 = 50{,}000$ seconds.

7. A classification model has precision = 0.80 and recall = 0.60. What is the F1-score?
a) 0.65
b) 0.70
c) 0.75
d) 0.80
Answer: b) 0.70
Formula: $F1 = 2 \{(Precision \times Recall)/(Precision + Recall)\} = 0.70$

8. Which of the following hyperparameters does not directly control the number of splits in trees of a Random Forest?
a) max_depth
b) min_samples_split
c) max_features
d) n_estimators
Answer: d) n_estimators
Explanation: In Random Forest, n_estimators controls the number of decision trees (or "estimators") used to create the forest, and generally, a higher number of trees improves model performance and stability, but also increases computational cost. It controls the number of trees, not the splits within trees.

9. Consider we are performing Random Forests based on classification data, and the relative frequencies of the class you are observing in the dataset are 0.65, 0.35, 0.29

and 0.5. What is the Gini index?
a) 0.423
b) 0.084
c) 0.12
d) 0.25
Answer: c) 0.12
Explanation: We know Gini index = 1 – ∑ci=1$(p_i)^2$ where p_i represents the relative frequency of the class you are observing in the dataset and c represents the number of classes. And we have $p_1 = 0.65$, $p_2 = 0.35$, $p_3 = 0.29$ and $p_4 = 0.5$
Gini index = $1 – (0.65^2 + 0.35^2 + 0.29^2 + 0.5^2)$
$= 1 – (0.423 + 0.123 + 0.084 + 0.25)$
$= 1 – 0.88$
$= 0.12$

10. What technique does Random Forest use to create diverse decision trees?
    a) Bootstrapping
    b) Cross-validation
    c) Gradient boosting
    d) K-means clustering

    Answer: a) Bootstrapping

    Explanation: Random Forest uses bagging (bootstrap aggregating) and random feature selection to create diverse decision trees.